Solar Na appendix

Sun · Na · downloadable appendix

Document	Solar Na appendix
Subject	Solar photospheric abundance appendix · Na
Author	Ryan Schmitt
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Solar sodium: 6.264 ± 0.119

Asplund, Amarsi & Grevesse 2021: 6.22 ± 0.03 · this measurement +0.044 dex

The three-dimensional non-LTE products apply published line-by-line corrections (Asplund et al. 2021; Canocchi et al. 2026) to our one-dimensional fits; they are not our own three-dimensional calculation. The H- and K-band products sit about 0.4 dex below the visible ones. They are shown as measured, with their full error bars.

Our best HARPS product, chosen the same way as for iron: the Reference Grade analysis with the most complete treatment, then the most lines. HARPS is the instrument we use for the stars that follow, so the Sun is anchored on it. Products and engines are not averaged together.

VIS: 6.264 ± 0.119 · HARPS · VIS · Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

Highlighted Products

near-UV

6.671 ± 0.163

±0.136 stat · ±0.089 syst

Na I · Deep-graded lines · 1D NLTE (Gerber) · Kitt Peak (Kurucz 2005) · Deep Grade

VIS

6.291 ± 0.059

±0.029 stat · ±0.051 syst

Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

6.338 ± 0.312

±0.083 stat · ±0.301 syst

Na I · Codex-graded lines · 1D NLTE (Gerber) · Kitt Peak (1984, molecfit) · Codex Grade

NIR

6.328 ± 0.052

±0.020 stat · ±0.048 syst

Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

5.779 ± 0.882

±0.692 stat · ±0.547 syst

Na I · Codex-graded lines · 1D NLTE (Amarsi) · CRIRES+ H · Codex Grade

K

5.903 ± 0.182

±0.132 stat · ±0.126 syst

Na I · Codex-graded lines · 1D NLTE (Amarsi) · CRIRES+ · Codex Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	30 products -- Na I · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	54 products -- Na I · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	28 products -- Na I · Deep-graded lines

Error-bar Forest Plot

near-UV — Kitt Peak (Kurucz 2005)
A(Na) dex; axis 6.15–6.88

Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.689 ±0.148 stat ±0.098 syst
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 2 lines		6.574 ±0.147 stat ±0.114 syst
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.671 ±0.136 stat ±0.089 syst
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.713 ±0.140 stat ±0.096 syst

near-UV — Kitt Peak (1984, molecfit)
A(Na) dex; axis 6.15–6.88

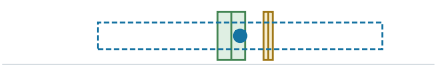
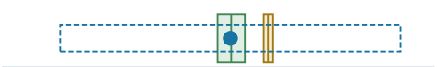
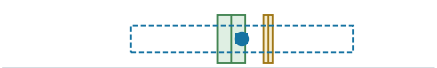
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.465 ±0.168 stat ±0.090 syst
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 2 lines		6.350 ±0.168 stat ±0.107 syst

near-UV — Kitt Peak (1984, molecfit)
A(Na) dex; axis 6.15–6.88

Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.437 ±0.158 stat ±0.098 syst
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.489 ±0.157 stat ±0.087 syst

near-UV axis: 6.15 · 6.34 · 6.52 · 6.70 · 6.88 A(Na) dex

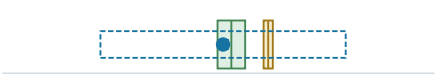
VIS — HARPS
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		6.278 ±0.105 stat ±0.043 syst
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		6.264 ±0.105 stat ±0.055 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 4 lines		6.226 ±0.108 stat ±0.053 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 4 lines		6.259 ±0.106 stat ±0.043 syst
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 4 lines		6.282 ±0.105 stat ±0.043 syst
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.228 ±0.013 stat ±0.292 syst
Na I · Deep-graded lines · 3D NLTE (Amarsi) Deep Grade · 2 lines		6.239 ±0.011 stat ±0.309 syst
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		6.102 ±0.018 stat ±0.175 syst
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.218 ±0.013 stat ±0.369 syst
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.243 ±0.014 stat ±0.241 syst
Na I · Codex-graded lines · 1D LTE Codex Grade · 22 lines		6.370 ±0.074 stat ±0.020 syst
Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		6.264 ±0.073 stat ±0.059 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 22 lines		6.295 ±0.077 stat ±0.036 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 20 lines		6.320 ±0.065 stat ±0.015 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 22 lines		6.380 ±0.097 stat ±0.017 syst

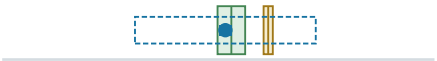
VIS — IAG FTS
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		6.394 ±0.026 stat ±0.060 syst
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		6.386 ±0.026 stat ±0.068 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		6.362 ±0.026 stat ±0.067 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		6.391 ±0.029 stat ±0.059 syst
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		6.401 ±0.026 stat ±0.060 syst
Na I · Codex-graded lines · 1D LTE Codex Grade · 10 lines		6.354 ±0.134 stat ±0.027 syst
Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 1 line		6.386 ±0.026 stat ±0.068 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 10 lines		6.318 ±0.136 stat ±0.041 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 10 lines		6.347 ±0.125 stat ±0.028 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 10 lines		6.353 ±0.126 stat ±0.025 syst

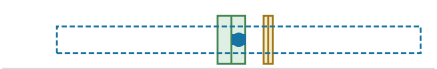
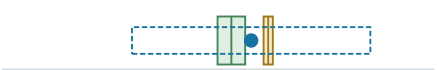
VIS — IAG FTS
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		6.293 ±0.023 stat ±0.044 syst
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		6.281 ±0.021 stat ±0.056 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		6.246 ±0.016 stat ±0.053 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 3 lines		6.277 ±0.018 stat ±0.042 syst
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 3 lines		6.298 ±0.022 stat ±0.043 syst
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.191 ±0.014 stat ±0.246 syst
Na I · Deep-graded lines · 3D NLTE (Amarsi) Deep Grade · 2 lines		6.202 ±0.012 stat ±0.266 syst
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		6.066 ±0.012 stat ±0.159 syst

VIS — IAG FTS
A(Na) dex; axis 5.72–6.66

Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.139 ±0.000 stat ±0.387 syst
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.207 ±0.014 stat ±0.196 syst
Na I · Codex-graded lines · 1D LTE Codex Grade · 6 lines		6.288 ±0.027 stat ±0.047 syst
Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 5 lines		6.281 ±0.034 stat ±0.060 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 6 lines		6.238 ±0.021 stat ±0.056 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 6 lines		6.272 ±0.022 stat ±0.047 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 6 lines		6.292 ±0.026 stat ±0.047 syst

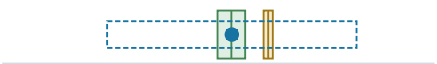
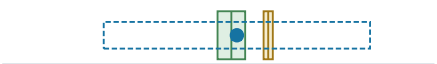
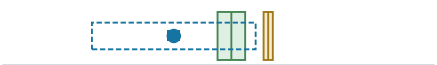
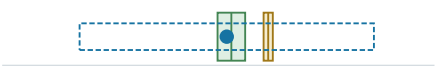
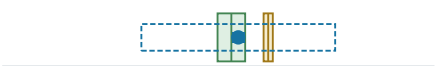
VIS — Kitt Peak (Kurucz 2005)
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		6.300 ±0.028 stat ±0.041 syst
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		6.287 ±0.029 stat ±0.053 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 4 lines		6.248 ±0.030 stat ±0.050 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 4 lines		6.284 ±0.030 stat ±0.021 syst
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 4 lines		6.305 ±0.029 stat ±0.039 syst
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.250 ±0.006 stat ±0.324 syst
Na I · Deep-graded lines · 3D NLTE (Amarsi) Deep Grade · 2 lines		6.260 ±0.008 stat ±0.340 syst
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		6.104 ±0.017 stat ±0.181 syst
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.236 ±0.015 stat ±0.395 syst
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.263 ±0.006 stat ±0.259 syst
Na I · Codex-graded lines · 1D LTE Codex Grade · 18 lines		6.300 ±0.076 stat ±0.019 syst

VIS — Kitt Peak (Kurucz 2005)
A(Na) dex; axis 5.72–6.66

Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		6.287 ±0.033 stat ±0.054 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 18 lines		6.248 ±0.077 stat ±0.036 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 18 lines		6.284 ±0.074 stat ±0.024 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 18 lines		6.305 ±0.074 stat ±0.019 syst

VIS — Kitt Peak (1984, molecfit)
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		6.304 ±0.029 stat ±0.039 syst
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		6.291 ±0.029 stat ±0.051 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 4 lines		6.252 ±0.030 stat ±0.049 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 4 lines		6.278 ±0.031 stat ±0.041 syst
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 4 lines		6.309 ±0.029 stat ±0.038 syst
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.221 ±0.013 stat ±0.271 syst
Na I · Deep-graded lines · 3D NLTE (Amarsi) Deep Grade · 2 lines		6.232 ±0.011 stat ±0.289 syst
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		6.095 ±0.013 stat ±0.178 syst
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.210 ±0.006 stat ±0.320 syst
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.235 ±0.014 stat ±0.210 syst
Na I · Codex-graded lines · 1D LTE Codex Grade · 17 lines		6.294 ±0.081 stat ±0.016 syst
Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		6.291 ±0.032 stat ±0.054 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 17 lines		6.247 ±0.082 stat ±0.034 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 16 lines		6.263 ±0.064 stat ±0.016 syst

VIS — Kitt Peak (1984, molecfit)
A(Na) dex; axis 5.72–6.66

Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 17 lines		6.298 ±0.076 stat ±0.016 syst

VIS axis: 5.72 · 5.96 · 6.19 · 6.43 · 6.66 A(Na) dex

red-optical — IAG FTS
A(Na) dex; axis 4.08–7.90

Na I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		6.327 ±0.160 stat ±0.422 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		5.990 ±0.259 stat ±1.884 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 4 lines		6.332 ±0.171 stat ±0.416 syst

red-optical — Kitt Peak (Kurucz 2005)
A(Na) dex; axis 4.08–7.90

Na I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		6.322 ±0.045 stat ±0.379 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.231 ±0.021 stat ±0.699 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 4 lines		6.271 ±0.073 stat ±0.411 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 4 lines		6.327 ±0.045 stat ±0.374 syst

red-optical — Kitt Peak (1984, molecfit)
A(Na) dex; axis 4.08–7.90

Na I · Codex-graded lines · 1D LTE Codex Grade · 5 lines		6.364 ±0.055 stat ±0.307 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		6.262 ±0.066 stat ±0.620 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		6.338 ±0.083 stat ±0.301 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 5 lines		6.370 ±0.055 stat ±0.295 syst

red-optical axis: 4.08 · 5.03 · 5.99 · 6.95 · 7.90 A(Na) dex

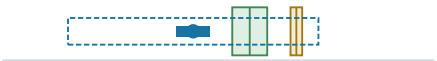
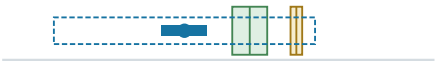
NIR — CRIRES+ J
A(Na) dex; axis 5.80–6.54

Na I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.213 ±0.292 stat ±0.203 syst

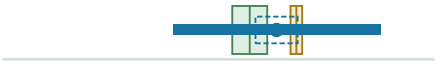
NIR — CRIRES+ J
A(Na) dex; axis 5.80–6.54

Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 1 line		5.901 ±0.076 stat ±0.053 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.196 ±0.279 stat ±0.227 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.215 ±0.292 stat ±0.191 syst

NIR — CRIRES+ Y (full arm)
A(Na) dex; axis 5.80–6.54

Na I · Codex-graded lines · 1D LTE Codex Grade · 1 line		6.123 ±0.029 stat ±0.215 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 1 line		6.105 ±0.029 stat ±0.217 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 1 line		6.108 ±0.039 stat ±0.224 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 1 line		6.122 ±0.029 stat ±0.215 syst

NIR — Kitt Peak (1984, molecfit)
A(Na) dex; axis 5.80–6.54

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		6.301 ±0.020 stat ±0.033 syst
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		6.328 ±0.020 stat ±0.048 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		6.295 ±0.020 stat ±0.045 syst
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		6.266 ±0.178 stat ±0.036 syst
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		6.303 ±0.020 stat ±0.032 syst
Na I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		6.252 ±0.147 stat ±0.070 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.067 ±0.167 stat ±0.055 syst
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.232 ±0.141 stat ±0.077 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		6.251 ±0.147 stat ±0.064 syst

NIR axis: 5.80 · 5.98 · 6.17 · 6.35 · 6.54 A(Na) dex

H — CRIRES+ H
A(Na) dex; axis 5.06–6.55

Na I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		5.796 ±0.692 stat ±0.546 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		5.779 ±0.692 stat ±0.547 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		5.803 ±0.713 stat ±0.593 syst

H axis: 5.06 · 5.43 · 5.80 · 6.17 · 6.55 A(Na) dex

K — CRIRES+
A(Na) dex; axis 5.73–6.34

Na I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		5.898 ±0.143 stat ±0.123 syst
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		5.903 ±0.132 stat ±0.126 syst
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 7 lines		5.899 ±0.142 stat ±0.119 syst

K axis: 5.73 · 5.88 · 6.03 · 6.19 · 6.34 A(Na) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every sodium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the sodium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Na)	Largest uncertainty terms (dex)
near-UV	Na I · Deep-graded lines	Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	6.350–6.671 (1D-NLTE)	fit / line-to-line scatter 0.14, microturbulence 0.08

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Na)	Largest uncertainty terms (dex)
VIS	Na I · Deep-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–2	6.202–6.260 (3D-NLTE)	fit window 0.25, model atmosphere 0.08
VIS	Na I · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–22	6.264–6.386 (3D-NLTE)	continuum placement 0.03, fit / line-to-line scatter 0.03
VIS	Na I · Asplund et al. 2021 lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–4	6.264–6.386 (3D-NLTE)	continuum placement 0.04, NLTE / 3D correction 0.03
red-optical	Na I · Codex-graded lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2–5	5.990–6.338 (1D-NLTE)	continuum placement 0.22, fit window 0.20
NIR	Na I · Codex-graded lines	CRIRES+ J, CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	1–3	5.901–6.232 (1D-NLTE)	fit / line-to-line scatter 0.14, fit window 0.07
NIR	Na I · Asplund et al. 2021 lines	Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE, 3D-NLTE	1	6.328 (3D-NLTE)	NLTE / 3D correction 0.03, fit / line-to-line scatter 0.02
H	Na I · Codex-graded lines	CRIRES+ H	1D-LTE, 1D-NLTE	3	5.779 (1D-NLTE)	fit / line-to-line scatter 0.69, telluric residual 0.54
K	Na I · Codex-graded lines	CRIRES+	1D-LTE, 1D-NLTE	7	5.903 (1D-NLTE)	fit / line-to-line scatter 0.13, telluric residual 0.12

Solar sodium context

Sodium is traced by neutral Na I lines from the near-ultraviolet to the K band. The reference lines are the five Asplund et al. 2021 used (4751, 5148, 6154, 6160 and 10746 Å), with the oscillator strengths they adopted. Na I departs from LTE, so each spectrum also carries non-LTE products; each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Na appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 27afcfcba7db6b4782ca756c1131fb9ddaa8dc3c
Feed Na.json v1.131 · updated 2026-10-27T15:01:19Z
Generated 2026-10-28T11:24:25-06:00 (source commit timestamp)
Feed SHA-256 0b09aec1b38fe26d0250d2e8bce185e8703d8398ceb3a180d70d87d33356a1ba
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Na comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. Na I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Na_Amarsi2020_PySME.csv). Sources: [5].

Model and grid sources. Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Model and grid sources. Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Solar abundance and indicator analyses. Published line set NA_AGSS21 (the line selection). Sources: [1].

Model and grid sources. Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte, A_1d_lte Sources: [1], [6].

Model and grid sources. Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte Sources: [1].

Product-to-source index

H · CRIRES+ H · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-fb210d1eee02

H · CRIRES+ H · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Na-products-a80b76db5d6a

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K · CRIRES+ · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-0552befecb14

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